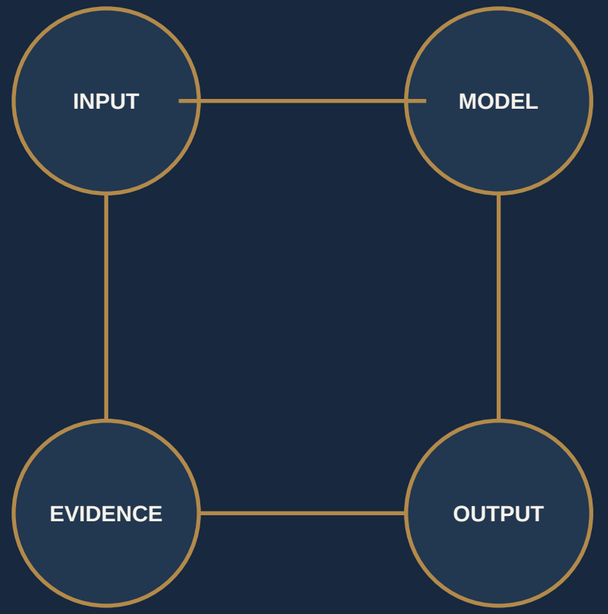
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robotic Perception

AI.202 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define how robots perceive their environments using cameras.
- Apply key methods including lidar.
- Build or analyze a practical system involving ultrasonic sensors.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

How robots perceive their environments using cameras

02

METHODS

LiDAR

03

SYSTEMS

Ultrasonic sensors

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · How robots perceive their env

How robots perceive their environments using cameras

02 · LiDAR

LiDAR

03 · Ultrasonic sensors

Ultrasonic sensors

04 · Radar

Radar

05 · IMUs

IMUs

06 · Encoders and tactile sensors

Encoders and tactile sensors

07 · Sensor models

Sensor models

08 · Calibration

Calibration

Core ideas, made explicit

AI.202 • Distinguish definitions, assumptions and evidence

- How robots perceive their environments using cameras
- LiDAR
- Ultrasonic sensors
- Radar
- IMUs

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.202 • Distinguish definitions, assumptions and evidence

- Encoders and tactile sensors
- Sensor models
- Calibration
- Filtering
- Feature extraction and sensor fusion

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

How robots perceive their environments using cameras

INPUTS

LiDAR

METHOD

Ultrasonic sensors

OUTPUT

Radar

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving how robots perceive their environments using cameras.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using how robots perceive their environments using cameras.
- Implement or simulate a baseline using lidar.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.